

NC Zoological Park: Integrated Geospatial & Regenerative Infrastructure Brief

Prepared by RHS AI Solutions | Candor, NC

1. Executive Summary & Operational Scope

The North Carolina Zoological Park (Asheboro Zoo) encompasses approximately 2,600 acres of rolling Piedmont topography, making it one of the largest zoological facilities in the United States. Managing this footprint presents unique operational challenges: high-traffic guest corridors that compact soil and degrade turf, animal habitat zones requiring precise hydrological control, and expansive stormwater dynamics driven by clay-heavy soils and seasonal rainfall intensity.

This brief outlines how RHS AI Solutions bridges the gap between macro-scale geospatial data and tactical, ground-truth implementation. While satellite-derived datasets provide baseline infrastructure mapping, our regional expertise in Piedmont hydrology, soil science, and regenerative landscape management translates that data into physical interventions — sub-surface drainage networks, soil health remediation, and sustainable turf transition programs.

2. Macro-to-Micro Integration: Beyond Global Asset Extraction

Macro-scale geospatial providers (such as Ecopia AI) deliver comprehensive vector datasets — building footprints, impervious surface boundaries, tree canopy coverage, and road networks — derived from high-resolution satellite and aerial imagery. These datasets provide the foundational canvas: a bird's-eye view of every structure, surface, and vegetation zone across the Zoo's 2,600-acre footprint.

However, satellite vectors alone cannot answer the critical questions facing facilities operations:

- Where is water pooling beneath the surface, and what subsurface flow paths drive it?
- Which zones are experiencing soil compaction beyond root-penetration thresholds?
- Where should French drains, bio-swales, or permeable paving be installed to prevent erosion and habitat waterlogging?
- How can turf management transition from synthetic chemical inputs to regenerative soil health without disrupting guest experience?

This is where RHS AI Solutions operates. We integrate macro-scale datasets with high-resolution ground-truth data — USGS 3DEP elevation models, SSURGO soil classifications, field-measured compaction readings, and localized rainfall patterns — to produce actionable hydrological models and regenerative maintenance protocols.

Positioning Statement: *Ecopia maps the canvas. RHS paints the picture — translating satellite vectors into sub-surface drainage plans, soil health programs, and regenerative turf transitions that protect both infrastructure and habitat.*

3. Piedmont Hydrological & Drainage Engineering

The Asheboro Zoo sits within the Carolina Slate Belt, characterized by clay-heavy ultisols with slow infiltration rates (0.1–0.6 in/hr) and high runoff coefficients. During intense Piedmont thunderstorm events, this creates three primary drainage challenges:

3.1 Surface Runoff Accumulation

Hardscaped exhibit zones and compacted guest corridors generate 3–5x more runoff than vegetated areas. Without engineered conveyance, this water concentrates at low points, causing erosion, sediment transport into animal habitats, and localized flooding.

3.2 Sub-Surface Waterlogging

Clay soils perched over shallow bedrock create perched water tables that saturate root zones for 48–72 hours after significant rainfall. This stresses native vegetation, destabilizes slope foundations, and creates breeding vectors for nuisance insects.

3.3 Engineered Solutions

Challenge	Conventional Approach	RHS Regenerative Solution
Surface Runoff	Concrete curbing, storm drains	Bio-swales with deep-rooted natives, permeable paving integration
Sub-Surface Waterlogging	Surface ditching, grading	Sub-surface French drain networks sized to soil percolation rates
Slope Erosion	Rip-rap, retaining walls	No-till warm-season grass establishment with check dam sequences
Exhibit Fringe Flooding	Structural berms, pumps	Integrated bio-retention cells with overflow spillways

4. Regenerative Turf & Habitat Maintenance Framework

The Zoo's 2,600 acres encompass multiple zone classifications, each with distinct maintenance challenges and regenerative opportunities:

Zone Classification	Primary Challenges	Conventional Maintenance	RHS Regenerative Protocol
High-Traffic Guest Corridors	Severe soil compaction, turf thinning	Chemical fertilization, constant resodding	Core aeration, compost tea inoculation, deep-root biological stimulation
Perimeter & Buffer Slopes	Sheet erosion, sediment runoff	Broad-spectrum herbicide application	Native warm-season grass establishment, no-till stabilization, check dams
Exhibit Fringe Areas	Standing water, localized stagnation	Surface ditching, structural grading	Integrated bio-retention cells, subsurface drainage networks
Habitat Buffer Zones	Invasive species encroachment	Mechanical removal, chemical spot-treatment	Prescribed native plant competition, biological pest control
Operational & Service Areas	Heavy equipment compaction, oil/contaminant runoff	Gravel resloping, containment ponds	Permeable hardscape conversion, phytoremediation strips

5. Phased Implementation Roadmap

- 1. Phase I: Baseline Topographical & Hydrological Audit:** High-resolution drone/ground asset mapping combined with flow-path modeling. Integrate existing satellite-derived vector datasets with USGS 3DEP elevation data to identify critical drainage bottlenecks, compaction zones, and erosion hotspots across the full 2,600-acre footprint.

2. **Phase II: Critical Drainage & Erosion Mitigation:** Targeted installation of subsurface water management systems in high-risk visitor and animal zones. Prioritize areas where surface runoff threatens infrastructure integrity or habitat health. Size all drainage elements to actual soil percolation rates and design storm events.
3. **Phase III: Biological Soil Transition:** Shift high-impact turf zones from synthetic chemical inputs to regenerative soil health management. Implement compost tea programs, native plant integration, and biological pest control to reduce long-term maintenance costs while improving ecological outcomes.

6. Technical Specifications & Deliverables

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# RHS AI Solutions – NC Zoo Asset Intelligence Schema

asset_profile:
  facility: "North Carolina Zoological Park"
  location: "Asheboro, NC (Randolph County)"
  total_acreage: 2600
  soil_classification: "Clay-heavy Ultisols (Carolina Slate Belt)"
  infiltration_rate: "0.1-0.6 in/hr"
  design_storm: "10-year, 24-hour (5.5 inches)"

hydrological_modeling:
  dem_source: "USGS 3DEP 1/3 arc-second"
  soil_data: "USDA SSURGO (c.hydgrp field)"
  rainfall_data: "NOAA Atlas 14, Piedmont NC"
  modeling_engine: "Custom Python (whitebox-tools, pysheds)"

deliverables:
  - zone_classification_map: "GeoJSON, 5 zone types"
  - drainage_risk_heatmap: "1m resolution raster"
  - french_drain_specifications: "Linear ft, depth, gravel spec per zone"
  - swale_dimensions: "Depth, width, length per contributing area"
  - turf_transition_schedule: "Multi-season, per zone"
  - soil_amendment_prescription: "Organic inputs per 100 sq ft"

reporting:
  format: "Automated PDF + interactive web dashboard"
  frequency: "Quarterly (post-storm event ad-hoc)"
  integration: "Supabase + PostGIS for spatial query"
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7. Conclusion & Next Steps

The North Carolina Zoological Park represents a unique convergence of public infrastructure, animal habitat management, and ecological stewardship. Managing 2,600 acres of Piedmont terrain demands more than satellite imagery — it requires ground-truth intelligence that translates geospatial data into physical interventions.

RHS AI Solutions proposes the following immediate next steps:

- **On-Site Consultation:** 90-minute walkthrough with Facilities Operations to identify priority zones and current pain points
- **Pilot Hydrological Audit:** Focused assessment of one high-risk zone (recommend: main guest corridor or primary exhibit fringe) to demonstrate modeling accuracy and intervention ROI
- **Data Integration Review:** Assessment of existing Zoo GIS assets and integration pathway with RHS modeling pipeline

We welcome the opportunity to discuss how our regional expertise in Piedmont hydrology and regenerative landscape management can support the NC Zoological Park's operational excellence.

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